# DBMS



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1. ER MODEL
Entity, Attribute, Relationship
1:1, 1:N, M:N
Super, Candidate, Primary, Foreign keys

2. NORMALIZATION
1NF: atomic
2NF: 1NF + no partial dep
3NF: 2NF + no transitive dep
BCNF: every determinant is CK

3. SQL
DDL: CREATE, ALTER, DROP
DML: INSERT, UPDATE, DELETE
DQL: SELECT

SELECT col FROM t
WHERE cond
GROUP BY col
HAVING cond
ORDER BY col;

Joins: INNER, LEFT, RIGHT, FULL

4. ACID
Atomicity, Consistency, Isolation, Durability

5. ISOLATION LEVELS
Read Uncommitted -> Serializable

6. LOCKING
Shared, Exclusive
2PL, Deadlock detection

7. INDEXING
B-Tree, B+ Tree, Hash, Bitmap
Clustered vs Non-clustered

8. QUERY OPT
Heuristic, Cost-based

EXPLAIN